



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

August 2023

22MCA203 – OPERATING SYSTEMS WITH LINUX

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

Unit – I

	Marks	BT*	CO*	PO*
1. a) Compare the User perspective and system perspective of an operating system.	6	L2	1	1
b) What are clustered systems? Explain.	6	L2	1	1
c) Explain the different operating system services.	8	L2	1	1
2. a) What is a Process Control Block(PCB)? Explain. Also Describe the different states of the processes with a diagram.	6	L2	1	1
b) Describe any two types of special purpose systems.	6	L2	1	1
c) Explain the different types of system calls.	8	L2	1	1

Unit – II

3. a) Consider the round robin scheduling with a time quantum 4 milliseconds and the set of 3 processes with its burst time.

Process	Burst Time
P ₁	24
P ₂	3
P ₃	3

Draw a Gantt Chart and Calculate average waiting time using Round Robin Scheduling Algorithm. Also find the average waiting time.	10	L3	2	2										
b) What is critical section problem? Explain the three requirements to solve the critical section problem.	5	L2	2	1										
c) Describe the concepts of semaphores with its types.	5	L2	2	1										
4. a) Consider the following set of processes and its burst time in milliseconds.														
<table><tr><td><u>Process</u></td><td><u>Burst Time</u></td></tr><tr><td>P₁</td><td>6</td></tr><tr><td>P₂</td><td>8</td></tr><tr><td>P₃</td><td>7</td></tr><tr><td>P₄</td><td>3</td></tr></table>	<u>Process</u>	<u>Burst Time</u>	P ₁	6	P ₂	8	P ₃	7	P ₄	3				
<u>Process</u>	<u>Burst Time</u>													
P ₁	6													
P ₂	8													
P ₃	7													
P ₄	3													
By using the Shortest Job First Scheduling (SJF), draw a Gantt Chart and Calculate Average waiting time. Also find the average waiting time in case of Non-preemptive SJF.	10	L3	2	2										
b) Explain the Peterson's solution to the critical section problem.	5	L2	2	1										
c) What are Monitors? Explain.	5	L2	2	1										

Unit – III

5. a) Explain Banker's algorithm with the aid of an example.	10	L2	3	1
b) Explain Contiguous memory allocation and Segmentation.	10	L2	3	1
6. a) How paging and swapping are utilized in memory management? Explain	10	L2	3	2
b) Let the page reference string be: 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1. Three frames are allocated to the program in main memory. Determine the number of page faults using:				
i) LRU page replacement policy				
ii) Optimal Page replacement policy.	10	L2	3	2

Unit – IV

7. a) Explain Absolute Pathnames and Relative Pathnames.	10	L2	4	1
b) What is file? Explain categories of file system.	10	L2	4	1
8. a) Explain basic file attributes in Unix.	10	L2	4	1
b) Write a note on i) Matching the dot, ii)Quoting.	10	L2	4	1

P.T.O

Unit – V

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|-----|-----|---|----|----|---|---|
| 9. | a) | Explain the mechanism of process creation. | 10 | L2 | 5 | 1 |
| | b) | How to Kill Processes with Signals? Explain. | 10 | L2 | 5 | 1 |
| 10. | a) | Discuss the various common environment variables. | 10 | L2 | 5 | 1 |
| | b) | Explain following essential shell programming concepts: | | | | |
| | i) | If conditional | | | | |
| | ii) | Using test and [] to evaluate expression. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
